# EIC and ePIC: the machine and the detector this programme is projected on

2026-08-27 · the specification behind reports 1, 2 and 4 · every row carries its source and its **status** — design value, requirement, measured, derived here, or unknown · figures from `evgen/scripts/eic_beam_figures.py`

Every projection in this programme rests on numbers that belong to the accelerator or to the detector, not to us. This note gathers them in one place, says where each comes from, and — more importantly — says what each one *is*: a design value in a conceptual design report, a requirement the detector must meet, something actually measured, or a gap we have filled with a stand-in. The lithium case is called out separately, because almost nothing about it is specified and the little that is has to be derived.

> **Three things a reader should take away.** (1) **Ion energies are set by speed, not rigidity.** The two rings must share a revolution period, so the hadron γ is fixed by the ring circumference and the magnets supply whatever rigidity that demands. The $^{6}$Li menu is **41, and 99–138 GeV/u**, with nothing between (§1). (2) **Almost nothing about a lithium beam is published.** Not the emittance, not the intensity, not β*, not a luminosity. What can be derived is derived in §4; what cannot is listed in §6 with an owner. (3) **This programme's detector model is partly stand-in.** The tracking resolution and the electron-ID curve in `polligen.reco` carry no published ePIC source, and §6 says so explicitly rather than letting them pass as specification.

## 1 · The collider, and why the ion energies are what they are

The EIC is two rings in the RHIC tunnel: the **Electron Storage Ring** at 5, 10 and 18 GeV, and the **Hadron Storage Ring** carrying protons to 275 GeV and ions to the same magnetic rigidity. ePIC sits at IP6; a second interaction region at IP8 is designed but not funded. The crossing angle is 25 mrad at IP6 and 35 mrad at IP8, recovered by crab cavities.

> **The constraint that sets the ion menu.** Colliding bunches requires the two rings to have the *same revolution period*. The electrons are ultrarelativistic at every energy, so their period is fixed; the hadrons must match it, and since the circumference is adjustable only within a narrow range, **the hadron β — hence γ — is quantised by the ring geometry**. The magnets then have to supply whatever rigidity that γ implies for the species in the ring, up to the 275 GeV proton rigidity of 917.3 T·m.
>
> So two different limits bind at two different ends. At the *top* the magnets bind and the ion is **rigidity-capped** at p/nucleon = 275 GeV × (Z/A). At the *bottom* the circumference binds and the ion is **γ-matched** — the same *speed* as the proton configuration, and therefore a different energy per nucleon than rigidity scaling would give.

**How the ring actually does it**, verbatim from EPIOS pp. 12–13: *"their revolution frequencies have to be equal. This synchronization is accomplished by applying a radial shift of up to ±20 mm in the arcs, which facilitates a range of the Lorentz factor of* **118 < γ < 293**. *To allow for even lower ion energies, a 'Blue' arc between IR12 and IR2 will be utilized as a bypass. The average radius of this arc is about 90 cm smaller than that of the corresponding 'Yellow' arc, which reduces the circumference of the HSR by roughly 90 cm. The resulting circumference then corresponds to a Lorentz factor of* **γ = 43.5**.*"* Those two numbers are exactly the species menu at *top* energies — gold at 110 GeV/u is γ = 118.1 and a 275 GeV proton is γ = 293.1 — so the window describes what the ion programme needs rather than a hard bound, and the shift is quoted as "up to".

> **A conflict this note records rather than resolves.** Yellow Report Table 10.1 runs **100 GeV protons**, i.e. γ = 106.6, which falls in *neither* stated window. The 41 GeV (γ = 43.7) and 275 GeV (γ = 293.1) anchors are both inside them, and Table 10.2's gold at 41 GeV/u (γ = 44.0) sits in the bypass window alongside the 41 GeV proton — which is the independent confirmation that ions are γ-matched. This programme stays anchored on the Yellow Report's three configurations, because every beam-parameter table is indexed by them, and `tools/consistency_check.py` flags the 100 GeV point so it stays visible. It is a question for C-AD, not a repository error. **Status: unresolved between two EIC documents.**

**The check that settles it.** Yellow Report Table 10.2 lists gold at **41 GeV/u** ($\gamma$ = 44.02) against the 41 GeV proton's $\gamma$ = 43.70 — equal to 0.7%, which is exactly the u versus $m_p$ mass difference. Rigidity scaling would have put gold at 41 × 79/197 = 16.4 GeV/u. It does not. The same rule reproduces the published ³He menu — *"41, and 100–183 GeV/nucleon"* — exactly.

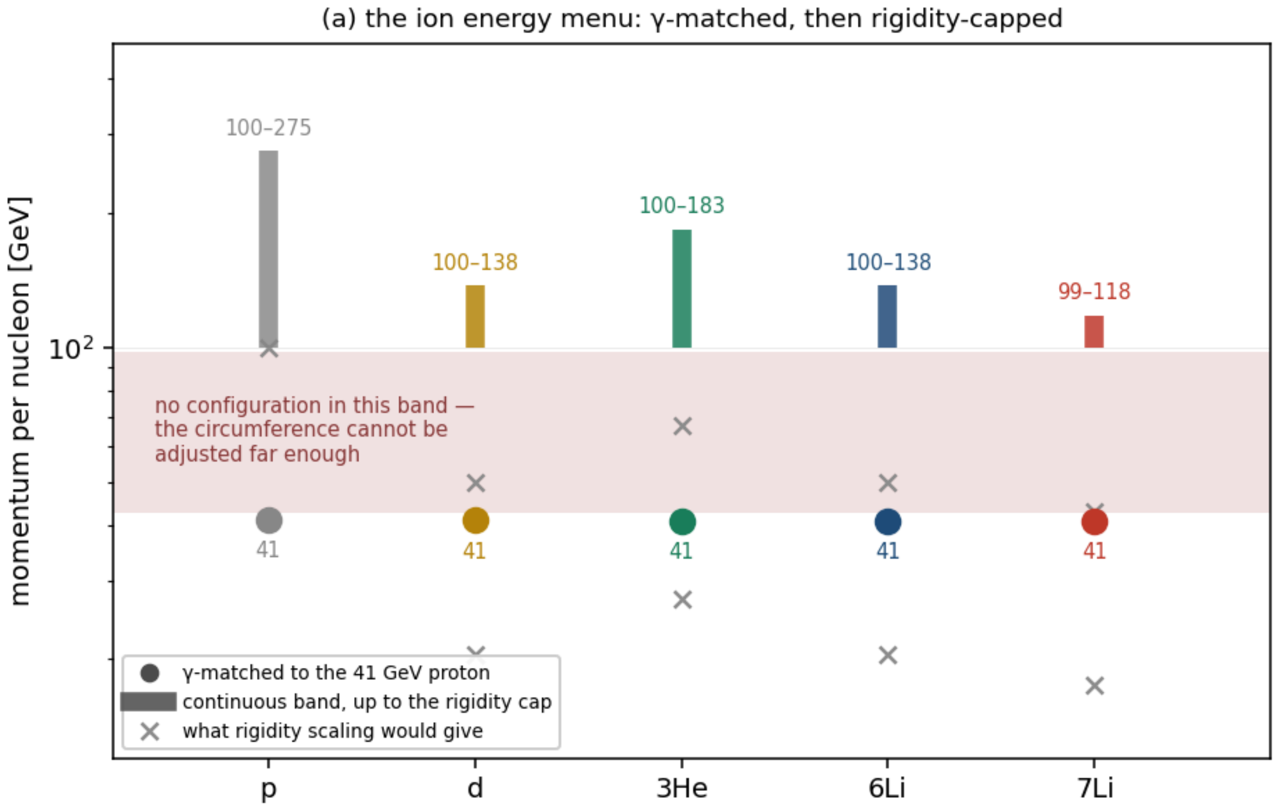


**Figure 1 · The ion energy menu.** Each species has an isolated γ-matched point at ≈41 GeV/u and a continuous band from ≈99 GeV/u up to its own rigidity cap — and nothing in between, because the circumference cannot be adjusted far enough. Crosses mark what rigidity scaling would have given, which is what this repository used until 2026-08-27. `eic_beam_figures.py`.

| SPECIES | A/Z | Γ-MATCHED POINT | CONTINUOUS BAND | RIGIDITY CAP | STATUS |
|---|---|---|---|---|---|
| p | 1 | 41 GeV | 100–275 GeV | 275 GeV | design value |
| d | 2 | 41.0 GeV/u | 100–137.5 | 137.5 | design value |
| ³He | 3/2 | 40.9 | 99.8–183.3 | 183.3 | design value (menu published) |
| **⁶Li** | 2 | **40.8** | **99.5–137.5** | 137.5 | **derived**; top ≈138 in EPIOS |
| **⁷Li** | 7/3 | **40.8** | **99.5–117.9** | 117.9 | **derived**; top ≈117 in EPIOS |
| Au | 2.49 | 41 | – (110 published) | 110.3 | design value |

Table 1 — `beams.default_configs` computes these; running the same machinery on a *proton* returns 41 / 100 / 275 exactly, which is the implementation's self-check. Only the lithium rows are derived rather than published, and only their top energies appear in EPIOS.

## 2 · The electron beam

| QUANTITY | VALUE | SOURCE | STATUS |
|---|---|---|---|
| energies | 5, 10, 18 GeV | YR §10.1 | design value |
| β* (h/v) | 45 / 5.6 cm | EPIOS Table I (EIC column) | design value |
| RMS emittance (h/v) | 20 / 1.3 nm | EPIOS Table I | design value |
| bunch charge | 28 nC | EPIOS Table I | design value |
| bunches | 1160 | EPIOS Table I | design value |

| | | | |
|---|---|---|---|
| divergence h/v, 18 GeV | 202 / 187 (HD), 89 / 82 μrad (HA) | YR Table 10.1 | design value |
| Δp/p | 5.8–10.9 $\times 10^{-4}$ | YR Table 10.1 | design value |
| polarization | 70% assumed in the YR physics studies | YR §10.1 | assumption for projections |

## 3 · The hadron beam

Yellow Report **Table 10.1** (e+p) and **10.2** (e+Au) are the authoritative divergence and momentum-spread tables, and they carry both optics configurations. The **high-divergence** and **high-acceptance** settings differ by a β* choice: the YR states explicitly that $\sigma \propto 1/\sqrt{\beta^*}$ and $\beta^* \propto 1/L$, so acceptance is bought with luminosity.

| PROTON CONFIG | HD H/V [MRAD] | HA H/V | ΔP/P [$10^{-4}$] | L [$10^{33}$ CM$^{-2}$S$^{-1}$] HD / HA |
|---|---|---|---|---|
| 275 GeV, 290 bunches | 150 / 150 | 65 / 65 | 6.8 | 1.54 / 0.32 |
| 275 GeV, 1160 bunches | 119 / 119 | 65 / 65 | 6.8 | 10.0 / 3.14 |
| 100 GeV | 220 / 220 | 180 / 180 | 9.7 | 4.48 / 3.14 |
| 100 GeV (vs e 5) | 206 / 206 | 180 / 180 | 9.7 | 3.68 / 2.92 |
| **41 GeV** | 220 / 380 | **220 / 380** | 10.3 | 0.44 / 0.44 |

Table 2 — YR Table 10.1, hadron rows, verbatim. **Status: design value.** Note the last row: at 41 GeV the two optics are *identical* — the only configuration with no separate high-acceptance option, and the one where this programme most needs one (§6, and report 4 §7).

| GOLD CONFIG | STRONG HADRON COOLING H/V | ΔP/P | STOCHASTIC COOLING H/V | ΔP/P |
|---|---|---|---|---|
| 110 GeV/u | 218 / 379 μrad | $6.2\times10^{-4}$ | 77 / 380 | $10\times10^{-4}$ |
| 41 GeV/u | 275 / 377 | $10\times10^{-4}$ | 174 / 302 | $13\times10^{-4}$ |

Table 3 — YR Table 10.2, gold rows. The two cooling scenarios differ by a factor 2.8 horizontally at 110 GeV/u, which is a reminder that the cooling choice is worth as much as the optics choice.

**Cooling.** The baseline is conventional electron cooling **at injection**, operationally demonstrated at LEReC. EPIOS p. 13 is explicit that store cooling is not in it: *"At a later stage, cooling at collision energies could be added"*, worth a factor two in average luminosity on a Coherent electron Cooling scheme. **So the emittance grows under intrabeam scattering through the fill with nothing fighting it**, and any argument that a low-intensity beam settles to a colder *equilibrium* has no equilibrium to settle to. **Status: design value; the store-cooling upgrade is explicitly later-stage.**

**Other hadron parameters** (EPIOS Table I, EIC column, design values): β* = 80 / 7.2 cm, RMS emittance 11 / 1 nm, bunch charge 11.0 nC, 1160 bunches, bunch length 6 cm, $\sqrt{s}$ = 105 GeV, peak luminosity $10^{37}$ cm$^{-2}$ s$^{-1}$. The beam is deliberately **flat**, $\varepsilon_x/\varepsilon_y \approx 11$, with β* matched to it so that the two *divergences* come out nearly equal — which is why Table 2's h and v agree at 275 and 100 GeV but not at 41.

## 4 · Lithium: published, derived, unknown

**What is published** is thin. EPIOS (arXiv:2510.10794, PRC 113:060501) gives the spin factors and the top energies and states the case; it gives *no* lithium luminosity, emittance or intensity, and describes the polarized-lithium source as early-stage. Hamwi & Hoffstaetter (arXiv:2509.18558, PRAB 29:073501) give the resonance spectrum.

| QUANTITY | $^{6}$LI | $^{7}$LI | SOURCE / STATUS |
|---|---|---|---|
| spin | 1 | 3/2 | nuclear data |
| A/Z | 2 | 7/3 | — |
| G factor | −0.178 (EPIOS), −0.1818 (Hamwi Table I) | +1.532 / +1.5196 | published; the two disagree in the last digits |
| max \|Gγ\|, resonances | 27, 81 | 191, 573 | Hamwi Table I, **design study** |
| snake scheme | deuteron-like: **not amenable to Siberian snakes** (small G needs impractically long snakes); partial | partial snakes | Hamwi; EPIOS |

| | | | |
|---|---|---|---|
| | solenoid + tune jumps | | |
| top energy | `137.5 GeV/u` | `117.9 GeV/u` | derived from the rigidity cap; ≈138 / ≈117 in EPIOS |
| energy menu | `41, and 99–138` | `41, and 99–118` | **derived** (§1) |
| σ_θ h/v, high acceptance | `220/380, 180/180, 92/92 µrad` | ≈same at the two lower | **derived** (below) |
| Δp/p | `7–10 ×10⁻⁴` | same | **derived**; species-insensitive |
| emittance, intensity, β*, luminosity | `— nothing published —` | | **unknown** (§6) |

### 4.1 · The divergence, derived

σ_θ = √(ε_N/(βγ·β*)), so at a given machine configuration — where β* is common to every species because the lattice is set by rigidity — the only species dependence is through βγ. And §1 says a **γ-matched** ion has the proton's βγ exactly. So:

$$\sigma_\theta(\text{ion}) = \sqrt{\frac{(\beta\gamma)_p}{(\beta\gamma)_{\text{ion}}}}\ \sigma_\theta(p)$$

which is **1** wherever the ion is γ-matched — the two lower configurations — and **√2** where it is rigidity-capped with A/Z = 2, i.e. ⁶Li at the top. So a ⁶Li carries the proton's divergence at 41 and 100 GeV and pays the √2 only at 18 × 275 — *not* a blanket factor, which is what an earlier draft of this programme's correction assumed.

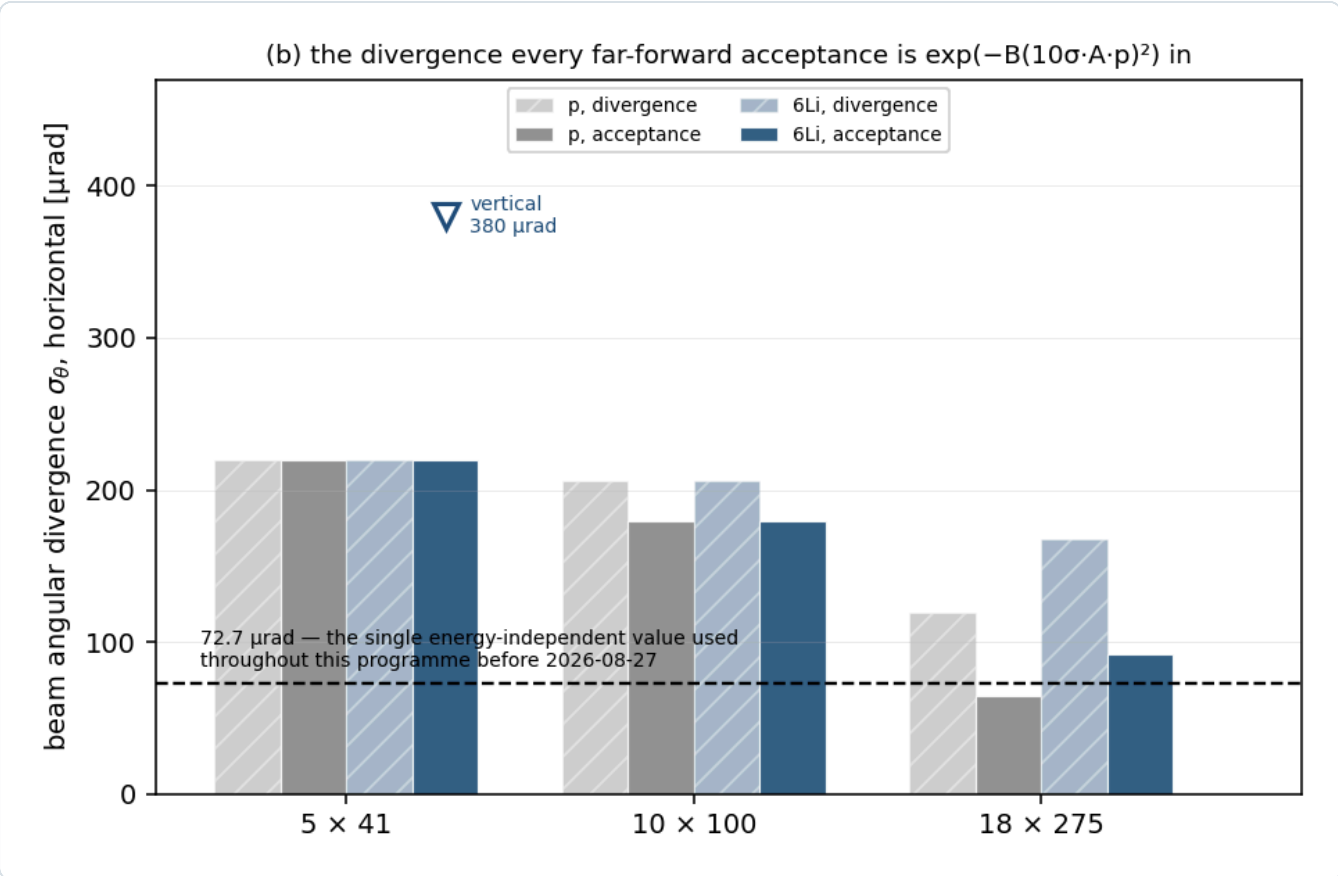


**Figure 2 · The divergence.** Protons and ⁶Li per configuration and optics. The species √2 appears only at 18 × 275, where ⁶Li is rigidity-capped; at the two lower configurations it is γ-matched and identical. The dashed line is the single energy-independent 72.7 µrad this programme used until 2026-08-27 — below almost everything. `eic_beam_figures.py`, `reco.sigma_theta_for`.

**Is equal ε_N defensible for lithium?** Gold is the published test. Scaling the 275 GeV proton to Au at 110 GeV/u by 1/√(βγ) alone predicts 236 µrad against an observed 218 (h) and 379 (v) — so ε_N(Au)/ε_N(p) = 0.85 horizontally and 2.6 vertically. Intrabeam scattering at fixed beam current goes as $Z^3/A^2$: **0.75 for ⁶Li**, 1 for a proton, ≈17× that for gold. RHIC deuterons showed no measurable IBS growth while gold grew 20–45%/h. Lithium is a proton-class ion, so equal ε_N is the right starting assumption — and it is an assumption, not a specification.

## 5 · The ePIC detector

### 5.1 · Far-forward — where this programme lives

| SUBSYSTEM | POSITION | ACCEPTANCE | TECHNOLOGY | STATUS |
|---|---|---|---|---|
| B0 | `z ≈ 5.4–6.4 m` | `5.5 < θ < 20 mrad` | AC-LGAD + EMCal | design value |
| Off-Momentum Detectors | `z ≈ 22.5, 24.5 m` | `θ < 5 mrad, x_L 0.45–0.65` | AC-LGAD | design value |
| Roman Pots | `z = 32.55, 34.25 m` | `θ < 5 mrad, x_L 0.6–0.95,` **`inner edge optics-dependent`** | AC-LGAD, in-vacuum | geometry file (current `main`) |
| ZDC | `z ≈ 37.5 m` | `θ < 4–5.5 mrad, neutrals` | W/Si + crystal | design value |

Table 4 — the Roman-Pot z positions are read from `compact/far_forward/roman_pots_eRD24_design.xml` on the current `main` branch and are **not** the 26 / 28 m the Yellow Report, the ePIC wiki and several papers still quote. The geometry moved; the documents did not.

**The Roman-Pot sensor and its readout.** AC-LGAD at **500 × 500 μm pitch** with a 100 × 100 μm metal electrode over a **30 μm active thickness**, bump-bonded to EICROC. Timing ≈35 ps per plane is a *requirement*; HPK AC-LGADs measure 20–35 ps in test beam. Spatial resolution ≈140 μm from the pitch alone; ≈20 μm has been shown with charge sharing, but ePIC's stated pot baseline is **binary 500 μm without charge sharing**. EICROC provides a 10-bit 25 ps time-of-arrival TDC and an **8-bit 40 MHz SAR ADC** for charge.

**The pot insertion is quantised.** Modules are 16 × 16 mm and the insertion is 2.4 cm + a per-energy offset (0.27 / 0.71 / 2.96 cm at 18 × 275 / 10 × 100 / 5 × 41), with the geometry file's own comment: *"These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."* The 1 mm aluminium RF shields are currently **commented out** — *"we don't know if we will even need it — it depends on the beam impedence + RF heating, etc. Oct. 2025"*.

**Rates and environment.** The ePIC Preliminary Design Report gives a few Hz per channel in normal running and *"30–50 kHz at ~10σ"* from beam halo, on STAR experience. In-vacuum cooling is ≈100 W per sensor layer. No published radiation fluence or dose for the pot station was found.

### 5.2 · Central detector

| QUANTITY | VALUE | SOURCE | STATUS |
|---|---|---|---|
| EMCal σ_E/E, backward (η < −2) | `2%/√E ⊕ 1%` | YR Fig. 8.3; matches the $PbWO_4$ crystal expectation | **requirement, correctly sourced** |
| … backward transition (−2 < η < −1) | `7%/√E ⊕ 1%` | YR Fig. 8.3 | **stochastic exact; constant 2× optimistic vs YR Table 3.1** |
| … barrel (\|η\| < 1) | `10%/√E ⊕ 1%` | YR Table 8.20 band, (10–12)%/√E ⊕ (1–3)% | **optimistic corner; constant 2-3× optimistic vs the ePIC BIC requirement** |
| … forward (η > 1) | `7%/√E ⊕ 1%` | — | **MIS-SOURCED — 7%/√E belongs to −2 < η < −1 only** |
| … noise term | absent | YR Table 3.1 has 1–2%/E | **missing; 2% at 1 GeV** |
| tracking σ_p/p | `√((a p)² + b²), a, b η-piecewise` | **YR Fig. 8.3 / Table 11.2**; ATHENA Table 4 quotes the same as "Requirements" | **requirement** — barrel and forward exact, endcap constants padded |
| tracking σ_θ | `5/3/2/1/2/3/5 mrad by η` | — | **PLACEHOLDER — no angular requirement in either YR table** |

| | | | |
|---|---|---|---|
| electron ID efficiency | `0.70–0.95 by η` | constructed between ATHENA JINST 17 P10019 T5 and ECCE NIM A 1055 168464 §3.5.2 | **anchored at one point only; the sources differ by ≈25 points in the barrel** |

Table 5 — fact-checked 2026-08-27. **An earlier version of this table called the tracking resolution a placeholder with no published source. That was wrong**: it is the Yellow Report requirement, matched exactly in the barrel and forward rows and padded conservatively in the endcaps. What follows is a caution rather than a reassurance — *agreement with a requirement is not validation*, since the table *is* the requirement. ePIC full simulation reaches 0.45–0.6% at p = 1 GeV/c, so the barrel smearing is realistic to about ±20%.

> **Where the stand-ins actually bite, and where they do not.** Every sweet spot this programme publishes puts the scattered electron **backward** (η = −2.9 to −1.6), where the calorimeter table is right to within 20% and where `RecoModel(energy="best")` hands the event to the tracker anyway. The mis-sourced forward row is latent, not active.
>
> The electron-ID curve almost certainly does not matter at all: the tensor observable is a **spin-state ratio**, and the estimator's own derivation shows that any efficiency common to all fills *cancels bin by bin*. $\varepsilon_{eID}$ should enter the statistical error and nothing else.
>
> The angular resolution is the one to watch, and the fix is not what it looks like. The 0.1 mrad that appears in some ePIC talks is a toy smearing input, superseded three slides later in the same talk by a full-simulation $\sigma_\theta = 72/p_T \oplus 2.8$ mrad — which at DIS $p_T$ is *worse* than the 3 mrad placeholder. And `smear_electron` carries no separate beam-divergence term on θ′, so the placeholder is the model's only stand-in for the irreducible IP divergence floor (81–211 μrad, which the YR says "cannot be corrected on an event-by-event basis"). Replacing 3 mrad with 0.1 mrad would *delete* that floor. The honest bracket is a factor ≲ 3.

## 6 · What this programme assumes that the specification does not give it

Ranked by how much a wrong answer would move a published number.

| # | WHAT IS MISSING | WHAT WE DO INSTEAD | OWNER |
|---|---|---|---|
| 1 | **Lithium beam divergence and Δp/p**, per configuration and cooling scenario | derive from the proton tables with the γ-matched species step (§4.1) | C-AD / ePIC FF WG |
| 2 | **Is there a high-acceptance optics at 41 GeV?** The YR's two rows are identical there | assume not; report 4 §7 shows the coherent channel depends on it | C-AD |
| 3 | **Lithium emittance, intensity, β* and luminosity** — nothing published | assume proton-class $\varepsilon_N$, defended by the gold calibration and the $Z^3/A^2$ IBS scaling | C-AD / ANL source group |
| 4 | **EICROC input charge dynamic range in fC**, and LGAD gain suppression at ~9 MIP | assume the charge is usable; report 4 §4.1 shows the far-forward Z-ID depends on it entirely | OMEGA-IJCLab / BNL AC-LGAD group |
| 5 | **Per-optics transfer matrix at the pot planes** — $R_{11}$, $R_{12}$, $R_{21}$, $R_{22}$, D, D′ | use $R_{12}$ = 30.6 m, measured at 18 × 275 only, and work in angle elsewhere | ePIC FF WG |
| 6 | **ePIC tracking resolution and electron-ID** — no published curves | η-piecewise stand-ins, flagged in the code and in Table 5 | ePIC |
| 7 | **Radiation environment at the Roman Pots** — no published fluence or dose | not modelled | ePIC FF WG |
| 8 | **Polarization-preservation scheme for $^6$Li**, which cannot use Siberian snakes | assume it works at the EPIOS source targets | C-AD / EPIOS |
| 9 | **No EIC document specifies a TENSOR-polarization scale uncertainty**, for any species. The 1% requirement is vector-only. | $P_{zz}$ is taken as known; every tensor amplitude scales with it | ANL source + BNL polarimetry |
| 10 | **Relative luminosity BETWEEN FILLS of different $P_{zz}$ is unspecified.** The $R < 10^{-4}$ requirement is on bunch-by-bunch relative luminosity *within* a fill. | the spin-sorted ratio estimator assumes it is controlled; an error enters through $\bar{P}$ as a bin-independent offset | EIC project / ePIC luminosity |

| | | | |
|---|---|---|---|
| 11 | **EICrecon has no light-ion far-forward configuration** — `MatrixTransferStatic` selects its transfer matrix by matching the *beam proton* PDG code, so a lithium beam matches nothing. | **Expected, and not a gap in the specification.** Lithium is not in the official programme yet; putting it there is what this development is for. This repository does its own transport meanwhile, and the deliverable is a beamline configuration and a transfer matrix that EICrecon can adopt. | **this programme** |
| 12 | **Roman-Pot commissioning is deferred past year 1 and conditional** — the far-forward subsystems "will only be operated once the beam conditions are fully under control". | projections assume the pots exist from the start | ePIC |
| 13 | **A 9 GeV electron beam is the early-science baseline**, and no ESR lattice, emittance, β* or polarization table exists for it. | only 5, 10 and 18 GeV are modelled | C-AD |

> **The honest summary.** The electron and proton beams are specified in detail and this programme uses them as such. But the list above separates into three kinds, and they call for different responses. Items 1–8 and 12–13 are **machine and detector inputs** — someone else's to supply, and the programme's job is to ask precisely. Items 9 and 10 — the tensor-polarization scale and the between-fill relative luminosity — are **ours to specify**, because they are properties of a measurement nobody has designed yet; §7 states them as this programme's own assumptions rather than leaving them blank. Item 11 is neither: lithium is absent from the official reconstruction chain because it is absent from the official programme, and **supplying it is the point of this development**. The *ion* beam is specified for gold and, thinly, for $^3$He; for lithium essentially nothing exists beyond a top energy and a G factor, and §4 is a derivation, not a citation. The detector is specified where it matters most for us — the far-forward geometry is in a file we can read — and least where the central reconstruction lives, which is why Table 5 carries two rows marked PLACEHOLDER rather than a number that would look like a specification.

## References


[1] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (split across `refs/2103.05419_part1.pdf` … `part4.pdf`). Tables 10.1 and 10.2 (beam parameters per configuration and optics), Table 11.48 (Roman-Pot p_T smearing), §11.6 (far-forward), §14.5.3 (superconducting nanowires), and the detector requirement tables.

[2] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794 (`refs/2510.10794.pdf`). Table I (HERA vs EIC parameters); the species table with G and mc²/G; the polarized-lithium case.

[3] E. Hamwi, G. H. Hoffstaetter, *Polarization Transmission in the Electron-Ion Collider's Hadron Storage Ring*, Phys. Rev. Accel. Beams 29 (2026) 073501, arXiv:2509.18558 (`refs/2509.18558.pdf`). Table I, the species-dependent resonance spectrum; the statement that small-G ions cannot use Siberian snakes.

[4] `eic/epic`, branch `main`: `compact/far_forward/roman_pots_eRD24_design.xml` and `compact/fields/beamline_*.xml`, read directly 2026-08-26. Station positions, module size, per-energy insertions, the RF-shield comment.

[5] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (`refs/ePIC_far_forward_talk_DIS_2023_v2.pdf`), and the ePIC Collaboration Meeting, JLab, July 2025. Acceptances, the AC-LGAD specification, the in-vacuum cooling load, the x-moving layer.

[6] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (`refs/2108.08314.pdf`). Table I, the far-forward acceptance windows this programme's router uses.

[7] ePIC Collaboration, *Preliminary Design Report*, September 2024. Roman-Pot rates: a few Hz per channel in normal running, 30–50 kHz at ~10σ from beam halo.


## Appendix A · Revision history

| DATE | REVISION |
|---|---|
| 2026-08-27 (later) | §1 gains the synchronisation mechanism verbatim from EPIOS — the ±20 mm radial shift covering $118 < \gamma < 293$ and the "Blue" arc bypass at $\gamma = 43.5$ — together with the conflict that the Yellow Report's 100 GeV proton ($\gamma = 106.6$) falls in neither window. §3 gains the finding that store cooling is *not* in the baseline. §5 Table 5 is fact-checked against the Yellow Report and the correction runs against this note's own earlier claim: **the tracking resolution is not a placeholder, it is the YR requirement**, which makes agreement with it a tautology rather than a validation. The calorimeter forward row is mis-sourced, its constant terms are 2–3× optimistic, and its |

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|---|---|
| | noise term is missing. §6 gains five further unknowns, of which three bear directly on this measurement rather than on the machine. |
| 2026-08-27 | First version. Written after the γ-matching correction (plans/10) made it clear that the programme had been carrying an accelerator assumption — that ion energies scale with rigidity — that no document supports and that the published gold and $^3$He menus refute. The beam sections are the corrected picture; the detector sections separate requirement from placeholder; §6 is the list of what we are assuming on the machine's and the detector's behalf. |

Status vocabulary: **design value** — appears in a design document as a parameter; **requirement** — a specification the subsystem must meet, not a demonstrated performance; **measured** — test beam or operations; **derived** — computed here from published inputs, with the derivation shown; **unknown** — not published anywhere we could reach. Beam divergences and Δp/p are Yellow Report Tables 10.1 and 10.2 unless stated. Ion energies are computed by `fastsim/polli_fastsim/beams.py` and pinned by `fastsim/tests/test_beams.py`. Built by `reports/build_report.py`; figures from `evgen/scripts/eic_beam_figures.py`.